

SIMULATION OF UNIVERSAL LOGIC BOARD USING VHDL

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Digital logic gate functions such as AND, OR, NOT, NAND, NOR, EX-OR, EX-NOR, etc., can be tested individually by using separate ICs designed for them. But it is neither cost-effective nor convenient to test each of these functions using



It consists of 16 input lines for 0-to-15 decimal values and four lines for 4-bit binary code. A 4-to-1 multiplexer with two select lines is also required. The

logic implemented in the program follows the block diagram as shown in Fig. 1. One can obtain the desired logic function from combinations of input lines using the 16-to-4 encoder. The multiplexer, which is a combinational digital logic circuit, connects any one of the input lines to its output lines, depending on its select

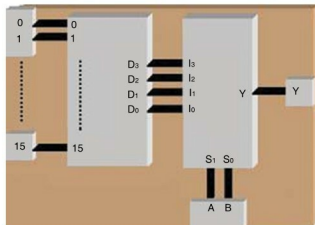


Fig. 1: Block diagram of universal logic board

separate ICs. Thus, in this article, we present a simple simulation program using Xilinx VHDL that verifies all the logic gate functions without using the ICs. The readers can find basics of VHDL, program design and testing procedures in EFY November 2009 issue under the project titled 'Simulation of time division multiplexer using VHDL.'

The design of universal logic board described in this project requires one decimal-to-binary encoder and a multiplexer. The program is designed in Windows XP environment. The encoder is a combinational digital logic circuit that provides an m-bit code on its output lines for an n-bit code on its input lines. The required encoder is 16-to-4 decimal-to-binary encoder.

input lines.

As shown in the block diagram, I_0, I_1, I_2 and I_3 are four input lines and S_0 and S_1 are the select lines for the multiplexer. The input lines A and B are connected to the select lines S_1 and S_0 , respectively. Y acts as the output for the logic gates. The output logic expression is given by:

$$Y = I_3 AB + I_2 AB' + I_1 A'B + I_0 A'B'$$

For instance, the output of AND function with input lines 1000 ($I_3 I_2 I_1 I_0$) is given by:

$$Y = AB + 0 + 0 + 0 = AB$$

Truth tables of various logic gates are shown in Table I through Table VIII. The outputs obtained from the truth tables will be used as input lines in the universal logic board design.

Binary codes obtained from the

TABLE I
AND

Inputs	Output
A B	Y
0 0	0
0 1	0
1 0	0
1 1	1

TABLE II
OR

Inputs	Output
A B	Y
0 0	0
0 1	1
1 0	1
1 1	1

TABLE III
NAND

Inputs	Output
A B	Y
0 0	1
0 1	1
1 0	1
1 1	0

TABLE IV
NOR

Inputs	Output
A B	Y
0 0	1
0 1	0
1 0	0
1 1	0

TABLE V
NOT A and B

Inputs	Output
A B	Y
0 *	1
1 *	0
* 0	1
* 1	0

TABLE VI
EX-OR

Inputs	Output
A B	Y
0 0	0
0 1	1
1 0	1
1 1	0

TABLE VII
EX-NOR

Inputs	Output
A B	Y
0 0	1
0 1	0
1 0	0
1 1	1

TABLE VIII
Buffer A and B

Inputs	Output
A B	Y
0 *	0
1 *	1
* 0	0
* 1	1

truth tables for various logic functions are given under Table IX. Decimal equivalents of each binary code are also listed in this table. The decimal equivalents and the binary codes are used in the program design to obtain waveforms of various logic gate functions.